WORKED SOLUTION

STEP I 1996, Question 8

Pure • Recurring Decimals

STEP I 1996, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q8

The key idea

A recurring block of digits is a geometric series in disguise: each repeat of the block is the previous one shifted k places to the right, so it is multiplied by 10^{-k}. Part (i) is the simplest case. In part (ii) we split the expansion into the part before the repetition starts (a finite decimal) and the recurring tail (a geometric series), and both pieces turn out to be integers divided by powers of ten or by $10^k - 1$.

Solution

Part (i)

The block 38 repeats every two places, so

$$\begin{aligned} 0.383838\ldots &= \frac{38}{100} + \frac{38}{100^2} + \frac{38}{100^3} + \cdots \\ &= \frac{38/100}{1 - 1/100} = \frac{38}{99}, \end{aligned}$$

summing the geometric series with ratio $\frac{1}{100}$, which is allowed because $|\frac{1}{100}| < 1$. Since $38 = 2 \times 19$ and $99 = 3^2 \times 11$ share no prime factor, this is in lowest terms:

$$0.383838\ldots = \frac{38}{99}.$$

Part (ii)

(The printed expansion $0 \cdot a_1 a_2 a_2 \cdots$ is a misprint for $0 \cdot a_1 a_2 a_3 \cdots$.) Each a_n is a digit from 0 to 9 and

$$x = \sum_{n=1}^{\infty} \frac{a_n}{10^n}.$$

We split this sum at $n = N$.

The first N digits. Let b be the integer whose decimal digits are $a_1 a_2 \ldots a_N$ (leading zeros allowed), that is,

$$b = \sum_{n=1}^{N} a_n 10^{N-n}.$$

Then

$$\sum_{n=1}^{N} \frac{a_n}{10^n} = \frac{1}{10^N} \sum_{n=1}^{N} a_n 10^{N-n} = \frac{b}{10^N}.$$

The tail. The condition $a_{n+k} = a_n$ for every $n > N$, applied repeatedly, gives $a_{N+mk+j} = a_{N+j}$ for every $m \geq 0$ and $1 \leq j \leq k$ (induction on m: the index $N + (m-1)k + j$ exceeds N, so we may add k to it). So after the Nth place the expansion consists of the block $a_{N+1} a_{N+2} \ldots a_{N+k}$ repeated for ever. Let c be the integer with these digits,

$$c = \sum_{j=1}^{k} a_{N+j} 10^{k-j}.$$

Group the tail into blocks of k terms; this is allowed because all the terms are non-negative and the series converges. The block for a given $m \geq 0$ is

$$\sum_{j=1}^{k} \frac{a_{N+mk+j}}{10^{N+mk+j}} = \frac{1}{10^{N+(m+1)k}} \sum_{j=1}^{k} a_{N+j} 10^{k-j} = \frac{c}{10^N} \left(\frac{1}{10^k} \right)^{m+1}.$$

? **Ask yourself** Why is each block exactly 10^{-k} times the one before it?

Moving from block m to block $m+1$ uses the same digits (by periodicity), but each sits k places further to the right. That divides each term by 10^k. This is the geometric series hidden in the recurring decimal.

Summing the geometric series with ratio $10^{-k} < 1$,

$$\sum_{n=N+1}^{\infty} \frac{a_n}{10^n} = \frac{c}{10^N} \cdot \frac{10^{-k}}{1 - 10^{-k}} = \frac{c}{10^N (10^k - 1)}.$$

Putting the pieces together. Adding the two parts,

$$x = \frac{b}{10^N} + \frac{c}{10^N (10^k - 1)},$$

where b is the integer formed by the first N digits $a_1 \ldots a_N$ and c is the integer formed by one period $a_{N+1} \ldots a_{N+k}$. This is the required form.

Rationality. Over a common denominator,

$$x = \frac{b(10^k - 1) + c}{10^N (10^k - 1)}.$$

The numerator is an integer, and the denominator is a positive integer because $k \geq 1$ gives $10^k - 1 \geq 9$. So x is a ratio of integers, which means x is rational.

Checks

- Part (i) is the case with no pre-period and $k = 2$, $c = 38$: the formula gives $38/(10^2 - 1) = 38/99$. (With $N = 1$ you get $b = 3$ and $c = 83$, and $\frac{3}{10} + \frac{83}{990} = \frac{380}{990} = \frac{38}{99}$, which agrees.)
- Take $0.12345345\ldots$, with $N = 2$, $k = 3$, $b = 12$, $c = 345$: $\frac{12}{100} + \frac{345}{100\times 999} = \frac{4111}{33300} = 0.1234534534\ldots$
- We also compared the formula with the summed expansion, in exact arithmetic, for 500 random choices of pre-period and repeating block; all agreed.

Common slips

- Forgetting the factor 10^{-N} on the tail: the repeating block starts at place $N + 1$, not at place 1.
- Using 10^k instead of $10^k - 1$ in the denominator: the first term of the geometric series is $c\,10^{-N-k}$, not $c\,10^{-N}$.
- Not checking lowest terms in (i): say why $\gcd(38, 99) = 1$.

Beyond STEP

The converse is also true: every rational number has an eventually recurring decimal expansion, because long division by q has only q possible remainders, so one must repeat. So the irrationals are exactly the reals whose expansions never settle into a cycle. The length of the period of $1/q$ is linked to the order of 10 modulo q, which is a first taste of group theory in number theory. Davenport's *The Higher Arithmetic* develops these ideas about congruences and periods beautifully.

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